

The Primon Gas at Criticality: the Fisher Computation, and the Connes–Consani Mechanism in Resistance Variables

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Status: Computation note. All identities in §2 were verified numerically (mpmath, 30 digits; agreement to truncation error where prime sums were cut off, and to 1e–31 where exact). The Connes–Consani digest in §5 is based on a direct reading of arXiv:2006.13771 (Connes & Consani, *Weil positivity and Trace formula, the archimedean place*, Selecta Math 27 (2021)); their sign convention differs from our Weil-note (they state RH as *negativity* of the geometric side, with $W_\infty = -W_\mathbb{R}$) — normalisation care required when the two notes are merged.

1. Setup

The **primon gas** (Julia; Bost–Connes — prior art): states are the integers $n \geq 1$, the energy of n is its resistance $E_n = \ln n$, Boltzmann weight $n^{-\beta}$, partition function $Z(\beta) = \zeta(\beta)$, defined as a state only for $\beta > 1$. By the Euler product the gas is a **free system of independent modes, one per prime**: $n = \prod_p p^{k_p}$, occupation k_p geometric with ratio $x_p = p^{-\beta}$, and the Gibbs measure factorises, $\pi(n) = \prod_p (1 - x_p) x_p^{k_p}$. Mode independence *is* the Euler product (Cathedral, Room of Density).

The question posed (open question 3 of the Weil note): compute the Fisher information form of this gas at the critical inverse temperature $\beta = \frac{1}{2}$ and test it against the shape requirements §5(i)–(v) of the Weil note — above all test (i): atomic masses $\ln p$, not $\ln n$.

2. Verified identities

(2a) Mean energy is the Weil atomic density.

$$U(\beta) = \langle \ln N \rangle_\beta = -\frac{\zeta'(\beta)}{\zeta(\beta)} = \sum_{n \geq 2} \Lambda(n) n^{-\beta} = \sum_p \ln p \cdot \langle k_p \rangle, \quad \langle k_p \rangle = \frac{x_p}{1 - x_p}.$$

Verified at $\beta = 2$: exact 0.569961...; via Λ -sum and via modes 0.569911... (prime cutoff 20,000; gap $\sim 5 \times 10^{-5}$ is the tail, $\approx 1/P$). **The masses $\ln p$ are forced with no choices**: they are the energy quanta of the modes. The positions $k \ln p$ are forced: they are the mode ladders (the nesting registers — the carry chains of the succession mechanism). The damping $e^{-E/2}$ is forced *iff* $\beta = \frac{1}{2}$. Read on the resistance line: **the prime-power resistance measure μ_P of the Weil note is the mode-resolved mean-energy density of the primon gas at the critical temperature**. Tests (i), (ii), (iii) pass — but this object is *linear* (a first cumulant), not the quadratic functional.

(2b) The Fisher information along the canonical family. Tilting temperature is the canonical one-parameter family; its Fisher information is the energy variance, i.e. the second cumulant:

$$I(\beta) = \text{Var}_\beta(\ln N) = \frac{d^2}{d\beta^2} \ln \zeta(\beta) = \sum_{n \geq 2} \Lambda(n) \ln n n^{-\beta} = \sum_p (\ln p)^2 \frac{x_p}{(1-x_p)^2}.$$

Verified at $\beta = 2$: all four expressions agree (0.8844818...; variance vs. cumulant identical to 10^{-31}).

(2c) Moments see everything; cumulants see only innovations. The raw second-moment measure of the gas has atoms at *every* $\ln n$ (mass $n^{-\beta}/\zeta(\beta)$ at E_n — including $n = 6, 10, 12, \dots$). The cumulant structure has atoms **only at prime powers** with Λ masses — composites drop out *because the modes are independent and cumulants are additive over independent components*. Verified in the atom table: at $n = 6, 10, 12$ the raw measure carries mass 0.0169, 0.0061, 0.0042 while the innovation measure carries exactly 0.

Observation 2.1 (the thermodynamic identity of the Möbius projection). (*Reading on the walls above.*) The passage from ζ to $\ln \zeta$ — the hinge — is the passage from moments to cumulants, and the innovation atoms of Observation 2.5 (Weil note) are precisely the **connected-correlation skeleton** of the primon gas. Composites carry no atom because a composite is a *product of independent things*: it has moments but no connected correlation. Paper 8’s Möbius/squarefree projection, which strips quantity and keeps the skeleton, is — thermodynamically — the moments→cumulants projection. This is the sharpest information-system identity of the framework’s central objects found so far: **innovation = connected correlation; the hinge = the cumulant functor**.

(2d) The Hagedorn wall. $Z(\beta) = \zeta(\beta)$ has its pole at $\beta = 1$; the series diverges for all $\beta \leq 1$, and $\zeta(\frac{1}{2}) = -1.4603\dots$ by continuation — negative, not a partition function. **The Gibbs state does not exist anywhere in the critical strip.** The critical line lies in the gas’s forbidden zone, two regimes past equilibrium. (Prior art: this is the Bost–Connes phase transition at $\beta = 1$; cite properly. The Cathedral’s reading of the strip — “exists only by analytic continuation, genuinely in-between” — is this wall seen from the analytic side.)

3. Verdicts on the shape tests

Candidate object	(i) masses $\ln p$	(ii) positions $\ln p^k$ only	(iii) damping $e^{-E/2}$	quadratic?	state exists at $\beta = \frac{1}{2}$?
Mean-energy density U	pass (forced)	pass (forced)	pass iff $\beta = \frac{1}{2}$	no — linear	no (formal only)
Fisher along $\beta (= \kappa_2)$	fail : masses $\Lambda(n) \cdot \ln n$	pass	pass iff $\beta = \frac{1}{2}$	yes, but in the wrong slot	no
Fisher of a probe-tilt $g(\ln N)$	fail	fail : atoms at all $\ln n$	—	yes	no

The precise failure of the Fisher form. Test (i) fails by *exactly one factor of $\ln n$* , and the reason is structural, not accidental: temperature couples to **energy**, energy is **position** on the resistance line, so every cumulant order multiplies the atoms by another factor of position. First

cumulant: masses Λ . Second cumulant (Fisher): masses $\Lambda \cdot \ln n$. The Weil functional wants Λ masses *inside a quadratic form* — first-cumulant masses in a second-order object. No tilt of a static Gibbs family can produce that: in a static ensemble, order of the form and power of the position weight are locked together.

The probe-tilt failure is the semantic one. A probe of the *scalar* total energy $g(\ln N)$ sees atoms at every integer — it has collapsed the vector to its length, and the innovation skeleton is invisible to it. Only **mode-additive (compositional) observables** — those respecting the prime-coordinate decomposition — have cumulants on the skeleton. This is Paper 3’s magnitude/direction split appearing as a theorem about which measurements can see the atoms: *the innovation structure is visible only to probes that respect composition*.

4. Structural conclusion: the atoms must live in the lags

The decisive mismatch is not the extra $\ln n$; it is *where the atoms sit*. Every Gibbs-ensemble form above carries its atoms in the **positions** of a static measure. The Weil quadratic form

$$Q(g) = W(g \star g^*)$$

is quadratic in the probe but **linear in the arithmetic distribution**: W enters as a *kernel evaluated at the lag $E - E'$* . For Q to be a covariance — the Bochner/GNS target of the Weil note — the source must be a **stationary process in resistance**, whose autocovariance has spikes at lags $\ln 2, \ln 3, 2 \ln 2, \ln 5, \dots$ with masses Λ . Atoms in the lags, not in the positions: the process must be statistically self-correlated under *translation by prime-power resistances*, i.e. under **multiplicative dilation by prime powers**.

A stationary flow whose time is resistance and whose self-correlations sit at prime dilations: that is, verbatim, the **scaling action** $\vartheta(\lambda) : f(x) \mapsto f(\lambda x)$ — translation by $\ln \lambda$ on the resistance line — which is exactly the flow on which the Connes program is built. The Fisher computation did not merely fail; it failed in the unique direction that *derives* why the operator approach uses the scaling flow rather than the Gibbs ensemble. The static gas supplies the right atoms, masses, and damping (2a) but in the wrong slot; criticality forbids its state outright (2d); the only repair consistent with the shape tests is to trade the ensemble for the flow and the measure’s positions for the covariance’s lags.

5. The Connes–Consani mechanism, read in resistance variables

Direct reading of arXiv:2006.13771. The correspondences are not loose:

(5a) Their coordinates are the resistance variables. Their working isomorphism is $w : L^2(\mathbb{R})^{\text{ev}} \rightarrow L^2(\mathbb{R}_+^*, d^*\lambda)$, $(w\xi)(\lambda) = \lambda^{1/2}\xi(\lambda)$, with $d^*\lambda = d\lambda/\lambda = dE$ — Haar measure on the multiplicative group is Lebesgue measure on the resistance line — and their automorphism $\Delta^{1/2} : f \mapsto x^{1/2}f$ exists, in their words, to replace the Mellin transform on the critical line by the plain Fourier transform. Absorbing the critical damping $e^{E/2}$ and working in (E, r) : that is Definition 2.2 of our Weil note. The framework’s variables are their variables.

(5b) The positivity root is a system trace — the GNS direction made concrete. Their functional $\text{Tr}(\vartheta(g) S \vartheta(g)^*)$ is **positive by construction** — it is the trace of a manifestly positive

operator, the scaling flow compressed onto Sonin space. Their Theorem 3: $\text{Tr}(\vartheta(f)S) = W_\infty(f) + \int f(\rho) \epsilon(\rho) d^*\rho$ — the archimedean Weil distribution is a compressed system trace, up to an explicit remainder; and their main theorem beats the remainder down (prolate functions, Toeplitz spectra, finitely many linear conditions, one rogue eigenvalue conditioned away, a constant $c \in (13, 17)$). This is Statement 4.3 of the Weil note executed at one place: exhibit W as (positive system functional) + (controlled correction). The system whose trace it is: the scaling flow — confirming §4 from the opposite direction.

(5c) The toolkit is Shannon’s. (*Wall on the identity of the tools; reading on the significance.*) The Sonin space is the orthogonal complement of $P_1 \vee \widehat{P}_1$ — the functions vanishing, together with their Fourier transforms, on $[-1, 1]$: the complement of a **finite phase-space cell**. The eigenbasis controlling the pair of cutoffs is the **prolate spheroidal wave functions** — the Slepian–Pollak–Landau apparatus, built at Bell Labs to count the degrees of freedom of a band-limited, time-limited channel; their “single quantum cell” is one phase-space information cell; the remainder is controlled by **Toeplitz** theory, the apparatus of stationary covariances. The deepest existing positivity result on the Weil functional is proven with the mathematics of channel capacity. For a framework whose claim is that the information layer is the projecting layer, this is the strongest external convergence on record — flagged as convergence, not as evidence.

(5d) The semilocal induction is literal, and the global gap is the all-at-once boundary. Their support conditions are lag windows on the resistance line: support in $(\frac{1}{2}, 2)$ is $|E| < \ln 2$ — no atoms touched, the pure archimedean case; the announced strategy for places $\{\infty, 2, 3, \dots, p\}$ uses support in (p^{-1}, p) — each adjoined prime widens the admissible window past its atom. Positivity place-by-place, the window growing one innovation at a time: the data-processing-shaped induction conjectured in the Weil note §6(b) is their explicit architecture. And the known obstruction of the whole program (cf. Deitmar’s semilocal note) is exactly the passage from *every finite set of places* to *all places at once*. (*Candle, explicitly flagged: this is the same shape as the Cathedral’s Seam — every finite truncation rational, the toll paid only at the infinite product. Whether the semilocal→global gap and the all-at-once toll are one phenomenon or a rhyme on the word “all” is an open candle/wall test, and nothing here banks it.*)

(5e) Their vanishing conditions are the pole terms. The required vanishing of $\widehat{g}$ at $\pm \frac{i}{2}$ removes the pole/existence terms — in the function-field analogy, focusing on H^1 — i.e. the contest is staged strictly between continuum and atoms once the primordial poles are subtracted. Consistent with the budget reading of the Weil note §3.

6. Honesty boundary

Verified: the identities of §2 (numerically, to stated precision; all are classical statistical mechanics of the free gas plus classical zeta identities — walls). Read directly from the source: the C–C statements of §5 (one paper, one reading; details of their Sections 4–6 not independently checked). Interpretive and marked: Observation 2.1 (innovation = connected correlation), the §4 lags-not-positions conclusion *as a heuristic derivation* of the scaling flow’s necessity (the implication “only repair consistent with the shape tests” is an argument, not a theorem), the Shannon-toolkit significance (5c), and the all-at-once candle (5d). Nothing in this note constrains the zeros.

7. Next steps

1. **Promote Observation 2.1.** “Innovation = connected correlation; Möbius projection = cumulant functor” is precise enough to state and prove as a short formal section (likely a lemma chain over independent geometric modes). If it survives, it belongs beside Paper 8 §2 as the thermodynamic identity of the two faces.
2. **The flow’s Fisher form.** Redo open question 3 for the *scaling flow* rather than the static gas: the natural family is now time-translation on the resistance line, and the candidate quadratic form is the flow’s spectral measure against $|h_g|^2$. The question becomes whether the semilocal trace formula *is* that Fisher form — which would close the loop between §4 and §5(b) in our own variables.
3. **Memory model.** The succession mechanism (NOT propagated through nesting) still lacks a memory model; the mode ladders $\{k \ln p\}$ used throughout §2 are exactly the registers it needs. A short note unifying “carry register = mode ladder = nesting chain” would let the genesis system and the gas share one formal vocabulary before the Turing-completeness question is attempted.
4. **Verification debts** carried forward: explicit-formula normalisation (Iwaniec–Kowalski), C–C sign conventions, Bost–Connes citation for the $\beta = 1$ transition, Knauf for the spin chain.

Classical content: Julia / Bost–Connes (primon gas, phase transition), standard cumulant theory, Connes–Consani 2021. The shape-test verdicts, Observation 2.1, the lags-not-positions argument, and the resistance-variable digest of §5 are the contribution, developed jointly by Alexander Pisani and Claude (Anthropic).